MES COLLEGE OF ENGINEERING KUTTIPPURAM

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

3-AXIS ROBOTIC ARM IMPLEMENTATION USING LOGIC ICs

A Subject Project Report

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CERTIFICATE

This is to certify that the subject project report entitled "3-AXIS ROBOTIC ARM IMPLEMENTATION USING LOGIC ICs", submitted by Hani Sakir KP (REDACTED), Lukman (REDACTED), Mohammed Shamil KK (REDACTED) and Nufail Ismail PP (REDACTED) to APJ Abdul Kalam Technological University, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering, is a record of the subject project work carried out by the students.

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ABSTRACT

This project develops a digital control system for a three-axis robotic arm using three 28BYJ-48 stepper motors and six pushbuttons. Each motor is controlled independently by one Forward and one Reverse button. An NE555 timer configured in astable mode generates a common adjustable clock signal. A 74HC14 Schmitt-trigger inverter conditions the pushbutton signals, while 74HC04 and 74HC08 logic gates implement direction selection and prevent Forward and Reverse commands from being active simultaneously. The submitted control-board schematic uses NOT and AND gate stages for direction selection and counter clock control.

For each axis, a 74HC193 up/down counter stores the commanded four-state sequence. The two least significant counter outputs are applied to one section of a 74HC139 dual 2-to-4 decoder. The decoder generates four active-LOW phase-selection signals, which are inverted as required before being applied to a ULN2003 driver stage. The ULN2003 switches the motor windings. The NE555 frequency can be adjusted to change the stepping rate and hence the motor speed.

The intended operation is continuous movement while a direction button is held, stopping when the button is released, and interlocking when both direction buttons are pressed together. The circuit is to be verified using LEDs at the driver inputs before connecting the motors. The controller is open-loop: the counter stores the commanded sequence state but does not measure the actual shaft position or detect missed steps.

Keywords: Robotic arm, digital logic, NE555, 74HC14, 74HC193, 74HC139, ULN2003, 28BYJ-48, stepper motor.

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Hani Sakir KP | Lukman | Mohammed Shamil KK | Nufail Ismail PP

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ABBREVIATIONS

Term	Meaning
CPU	Count-up clock input of 74HC193
CPD	Count-down clock input of 74HC193
FWD	Forward command
REV	Reverse command
CLK	Clock signal
RPM	Revolutions per minute
GND	Ground reference

CHAPTER 1

INTRODUCTION

1.1 Background

Robotic arms are used in automation, material handling, positioning and educational demonstrations. A multi-axis arm requires controlled movement of several joints. Stepper motors are useful in such systems because they rotate in discrete increments in response to electrical pulse sequences.

This project develops a three-axis robotic-arm controller using discrete digital logic ICs rather than a microcontroller. The design combines an adjustable clock generator, pushbutton input conditioning, direction-control logic, up/down counters, decoders and motor-driver stages.

1.2 Problem Statement

The project aims to provide independent Forward and Reverse movement for three stepper-motor axes. Each motor should continue stepping while its selected direction button is held, stop when the button is released, and not receive conflicting direction commands when both buttons are pressed simultaneously.

1.3 Objectives

- Generate a common adjustable clock using an NE555 timer in astable mode.
- Condition six pushbutton signals using 74HC14 Schmitt-trigger inputs.
- Implement direction selection and interlocking using NOT and AND logic.
- Use one 74HC193 up/down counter per motor axis.
- Decode the two least significant counter bits using 74HC139 sections.
- Drive three 28BYJ-48 stepper motors through ULN2003 driver stages.
- Verify the logic sequence with LEDs before connecting the motors.

1.4 Scope

The report covers the digital control and motor-interface logic for three axes. It does not include closed-loop position feedback, automatic homing, limit-switch protection, trajectory planning or mechanical design calculations. These can be considered as future improvements.

CHAPTER 2

EXISTING SYSTEM AND PROPOSED APPROACH

2.1 Existing Approaches

Stepper-motor controllers may be implemented using microcontrollers, dedicated stepper-driver ICs or discrete logic. Microcontroller-based systems offer programmable motion profiles and compact integration. Discrete logic provides a transparent implementation in which the clock, direction logic, counter state and decoder outputs can be observed directly.

2.2 Proposed Approach

The proposed controller uses one common NE555 clock and three independent axis-control channels. Each channel has two pushbuttons, input conditioning, direction-interlock logic, one 74HC193 counter, one 74HC139 decoder section, output inversion and one ULN2003 driver. Three decoder sections are required; two 74HC139 ICs provide four sections in total.

2.3 Design Considerations

• The 74HC193 inactive clock input must be held HIGH; no CMOS input should be left floating.

• The 74HC139 enable and outputs are active-LOW.

• The ULN2003 inputs are active-HIGH, so output polarity conversion is required.

• The exact 28BYJ-48 winding order must be checked against the motor wiring.

• The counter tracks the commanded phase state, not measured mechanical position.

CHAPTER 3

THEORETICAL BACKGROUND

3.1 NE555 Astable Multivibrator

The NE555 is configured in astable mode to generate a continuous square-wave clock. The timing capacitor charges and discharges through the external resistance network. A potentiometer in the timing path allows the clock frequency to be adjusted. The approximate frequency is:

Astable frequency equation:

$$f = \frac{1.44}{(R_A + 2R_B)C}$$

Where:

- f = Output frequency in Hz
- R_A, R_B = Timing resistances in ohms
- C = Timing capacitance in farads

3.2 74HC14 Schmitt-Trigger Inverter

The 74HC14 contains six Schmitt-trigger inverters. It is used to condition pushbutton signals and improve immunity to slow or noisy transitions. With a pull-up resistor and a normally-open switch to ground, the raw input is HIGH when released and LOW when pressed. The inverter output is active-HIGH when the button is pressed.

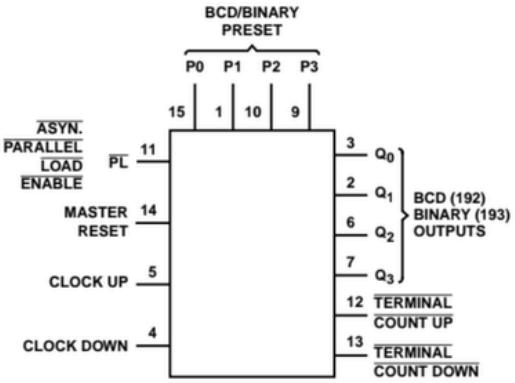
3.3 NOT and AND Logic

The 74HC04 provides NOT gates and the 74HC08 provides two-input AND gates. These gates implement mutually exclusive Forward and Reverse enables and clock selection. The submitted control-board schematic identifies 74HC04 and 74HC08 devices; it does not show a IC.

3.4 74HC193 Up/Down Counter

The 74HC193 is a synchronous 4-bit binary up/down counter with separate count-up and count-down clock inputs. In this design, Q0 and Q1 provide the four-state sequence 00, 01, 10 and 11. Q2 and Q3 are not used for this four-state sequence. The counter retains its commanded binary state when no count edge is applied; it does not detect the actual motor shaft position.

Functional Diagram



TRUTH TABLE

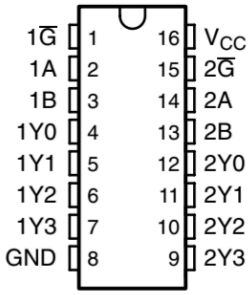
CLOCK UP	CLOCK DOWN	RESET	PARALLEL LOAD	FUNCTION
↑	H	L	H	Count Up
H	↑	L	H	Count Down
X	X	H	X	Reset
X	X	L	L	Load Preset Inputs

H = High Voltage Level, L = Low Voltage Level, X = Don't Care, ↑ = Transition from Low to High Level

3.5 74HC139 Decoder

The 74HC139 contains two independent 2-to-4 decoder sections. Each section has two address inputs, an active-LOW enable and four active-LOW outputs. One section is used for each motor axis. Two ICs provide four sections, leaving one section unused.

D OR PW PACKAGE
(TOP VIEW)



FUNCTION TABLE

INPUTS			OUTPUTS			
$\overline{G}$	SELECT					
	B	A	Y0	Y1	Y2	Y3
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	L	H	H	L	H	H
L	H	L	H	H	L	H
L	H	H	H	H	H	L

3.6 ULN2003 and 28BYJ-48

The ULN2003 is a seven-channel Darlington transistor array used as a low-side current-sinking driver. It interfaces the logic signals to the motor windings. The project uses the 5 V variant of the 28BYJ-48 stepper motor. The ULN2003 COM pin is connected to the positive motor supply to use the internal clamp diodes for inductive-load protection.

CHAPTER 4

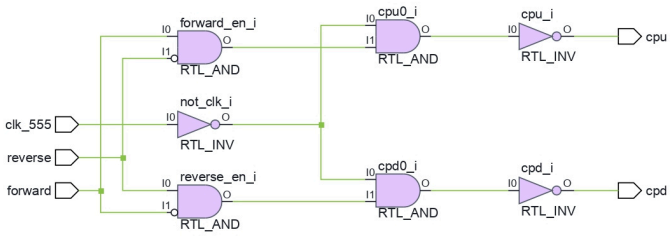
DESIGN AND CONTROL LOGIC

4.1 System Block Diagram

Stage	Signal flow / function
Clock generation	NE555 astable timer → common CLK
Input stage	Six pushbuttons → 74HC14 conditioning
Direction logic	74HC14 + 74HC04 NOT + 74HC08 AND → CPU / CPD
Counter stage	Three 74HC193 counters; one per axis
Decoder stage	74HC139 2-to-4 decoder section per axis
Output stage	74HC04 inversion → ULN2003 → 28BYJ-48 motors

The common clock is distributed to three independent control channels. Each axis has its own Forward/Reverse command pair, counter, decoder section and motor-driver channels.

4.2 Direction-Control



The 74HC193 counter generates a four-state binary sequence using its Q0 and Q1 outputs. These outputs are connected to a 74HC139 decoder, which selects the motor coil activation sequence.The same control arrangement is used for the other two motor axes.

$$F_{EN} = F \cdot \overline{R}$$

$$R_{EN} = R \cdot \overline{F}$$

$$CPU = \overline{(\overline{CLK} \cdot F_{EN})}$$

$$CPD = \overline{(\overline{CLK} \cdot R_{EN})}$$

4.3 Direction-Control Truth Table

FWD	REV	CPU	CPD	MODE
0	0	1	1	Stop
0	1	1	CLK	Reverse
1	0	CLK	1	Forward
1	1	1	1	Interlock (Lockout)

4.4 Four-State Full-Step Sequence

Step	Phase A	Phase B	Phase C	Phase D
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

4.5 Speed Estimate

Using the nominal 4096 full-step figure commonly quoted for the geared 28BYJ-48 output shaft, the approximate clock frequency is:

$$f = (4096 \times RPM) / 60$$

Approximate output speed	Approximate clock frequency
0.5 RPM	34 Hz
1 RPM	68 Hz
2 RPM	137 Hz
3 RPM	205 Hz
5 RPM	341 Hz

These are nominal estimates, not guaranteed operating speeds. Actual speed depends on the specific motor, gearbox, load, supply voltage and missed steps. The clock should be increased gradually during testing.

CHAPTER 5

HARDWARE IMPLEMENTATION

5.1 Input Stage

Each axis uses two normally-open pushbuttons. A 10 kΩ pull-up resistor connects each raw input node to +5 V, and the pushbutton connects the node to ground. The raw input is HIGH when released and LOW when pressed. The 74HC14 inverter produces an active-HIGH signal when pressed. The Schmitt-trigger action helps condition transitions; additional RC debounce may be used if required by testing.

5.2 Clock and Direction-Control Stage

The NE555 pin 3 output provides the common clock. NOT and AND gates form mutually exclusive Forward and Reverse enables. The NOT and AND gate network implements clock selection. The resulting CPU and CPD signals connect to pins 5 and 4, respectively, of the 74HC193.

5.3 74HC193 DIP-16 Connection Summary

Verify all pin numbers against the datasheet for the exact manufacturer and device variant before wiring. Do not leave any CMOS input floating.

5.4 Decoder and Output Stage

The two least significant counter outputs connect to the A and B address inputs of one 74HC139 decoder section. Its active-LOW enable is connected LOW. The four active-LOW decoder outputs are inverted before connecting to the ULN2003 inputs. The motor common wire connects to +5 V and the four winding terminals connect to the driver outputs, after confirming the specific motor connector pinout.

5.5 Power Supply and Construction

Use a regulated 5 V supply with adequate current capacity for the motors and logic. Connect logic and motor grounds together. Connect the ULN2003 COM pin to the positive motor supply. Place a 100 nF ceramic decoupling capacitor close to the supply pins of each logic IC and add bulk capacitance near the supply entry. Do not drive motor windings directly from 74HC outputs.

5.6 Recommended Implementation Order

- Test the NE555 clock independently at a low frequency.
- Verify the pushbutton signals and 74HC14 outputs.
- Verify direction interlock and CPU/CPD clock levels.
- Connect the 74HC193 and verify up/down counting.
- Connect the 74HC139 and verify the four decoded states.

- Test the inverter and ULN2003 input indicators with motors disconnected.
- Connect the motors only after confirming sequence, supply polarity and wiring.

CHAPTER 6

TESTING AND EXPECTED RESULTS

6.1 Verification Procedure

The circuit should be tested progressively before the motors are connected. Begin with a slow clock of approximately 1–2 Hz so that the counter and decoder states can be observed. LEDs with suitable current-limiting resistors can be used to monitor the four coil-control signals. The ULN2003 board indicator LEDs may also be used if their input polarity and circuit arrangement are understood.

6.2 Functional Test Plan

Test	Input condition	Expected observation
1	FWD=0, REV=0	CPU and CPD HIGH; counter holds state.
2	FWD=1, REV=0	CPU receives clock; counter increments.
3	FWD=0, REV=1	CPD receives clock; counter decrements.
4	FWD=1, REV=1	Both enables disabled; counter holds.
5	Decoder outputs	Four states appear in sequence, one selected at a time.
6	ULN2003 indicators	Driver input indicators follow the four phase commands.

6.3 Expected Result

The expected result is a three-axis robotic arm in which each 28BYJ-48 motor moves in the selected Forward or Reverse direction. Each motor is intended to run continuously while its direction button is held and stop when the button is released, retaining the last commanded sequence state. The common 555 clock frequency controls the stepping rate. The expected behavior must be confirmed by the staged tests and final motor operation.

6.4 Limitations

The controller is open-loop. The 74HC193 stores the commanded phase state but does not measure actual shaft position. If a motor stalls or misses steps under load, the counter state may differ from the mechanical position. LED sequencing verifies the logic path but does not prove that the motor has moved correctly. Final speed and torque limits must be established experimentally with the assembled arm.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

A three-axis robotic-arm controller using discrete digital logic ICs has been designed at the functional level. The architecture combines an NE555 astable clock, 74HC14 input conditioning, NOT/AND direction logic, 74HC193 up/down counters, 74HC139 decoders and ULN2003 motor drivers. The project demonstrates the practical use of multivibrators, combinational logic, counters and decoders in a multi-axis control application. A staged LED verification procedure is specified before connecting the motors.

7.2 Future Scope

• Add limit switches and a homing routine for safer travel limits.

• Add encoder or position-sensor feedback to detect missed steps and estimate actual position.

• Implement acceleration and deceleration to improve starting and stopping under load.

• Provide independent clock-frequency adjustment for each axis.

• Develop a PCB with short logic traces and separated motor-current paths.

• Investigate two-phase full-step or half-step drive if higher holding torque or finer positioning is required.

REFERENCES

[1] Texas Instruments, NE555 Timer datasheet. Use the exact manufacturer revision consulted for astable operation and electrical characteristics.

[2] Texas Instruments, SN74HC14 Hex Schmitt-Trigger Inverters datasheet.

[3] Texas Instruments, SN74HC04 Hex Inverters datasheet.

[4] Texas Instruments, SN74HC08 Quadruple 2-Input AND Gates datasheet.

[5] Texas Instruments, SN74HC193 4-Bit Synchronous Binary Up/Down Counter datasheet.

[6] Texas Instruments, SN74HC139 Dual 2-to-4 Line Decoder/Demultiplexer datasheet.

[7] Texas Instruments, ULN2003A Seven-Channel Darlington Transistor Array datasheet.

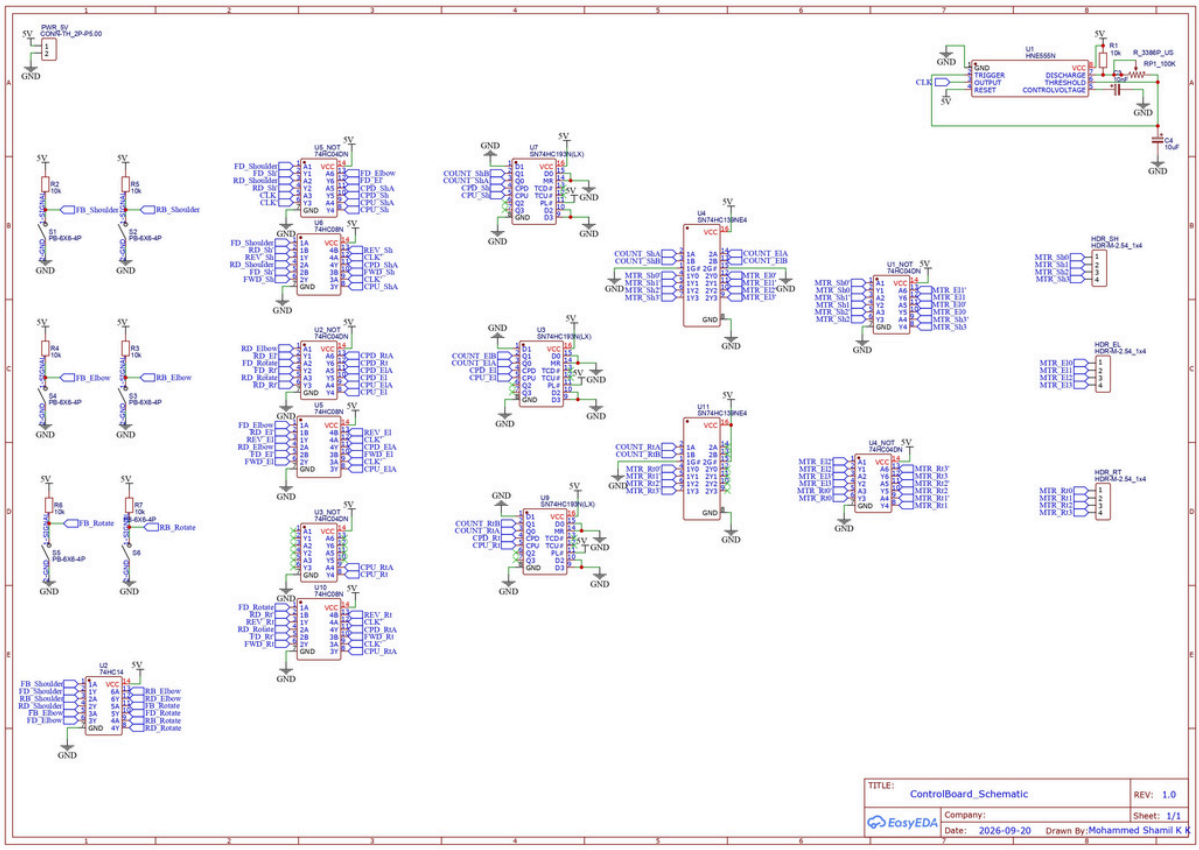
[8] 28BYJ-48 5 V stepper motor specification sheet for the exact motor variant used.

Data Sheets
https://drive.google.com/drive/folders/1CA9UgfUqsF2AB4I_VJeSis9Oav8bHMFR

APPENDIX

Appendix A – Final Control-Board Schematic

The following schematic is the submitted control-board design dated 20 September 2026. It shows the NE555 astable clock, six pushbutton inputs, 74HC14 input conditioning, 74HC04 inverter stages, 74HC08 direction-control logic, three 74HC193 counters, two 74HC139 decoder ICs and the three motor-output headers. The external ULN2003 driver boards and stepper motors connect at the motor-output interface; they are not drawn as part of this control-board sheet.



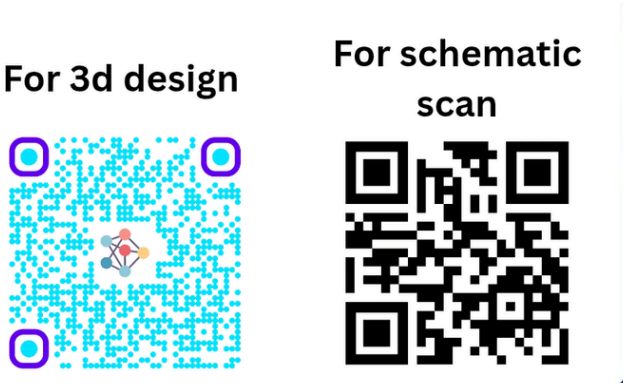


Figure A.1 – Submitted control-board schematic (source: uploaded schematic PDF).

Appendix B – Prototype Demonstration Photograph

The photograph below shows the project's laboratory demonstration setup, including the assembled logic-control prototype, connecting wires, pushbutton controls and indicator LEDs. The project poster is displayed behind the prototype. The photograph documents the controller-board demonstration; it does not establish measured motor performance or full-load operation of the completed robotic arm.

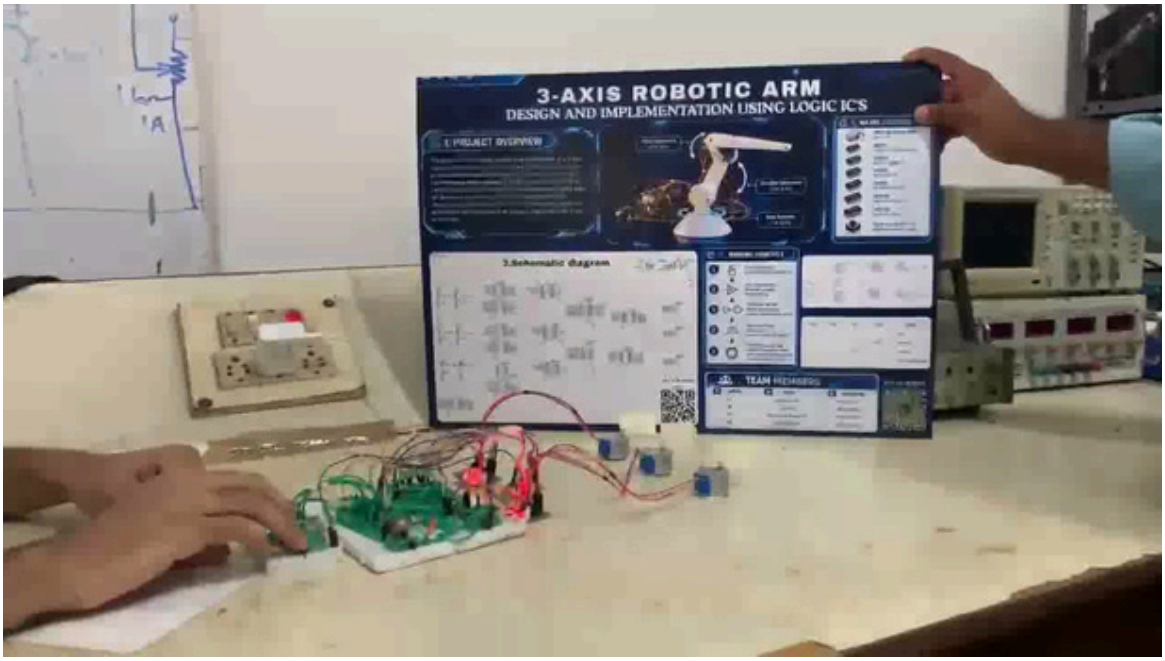


Figure B.1 – Laboratory demonstration photograph of the robotic-arm control-board prototype.

Appendix C – Bill of Materials

Component	Quantity	Remarks
NE555 timer	1	Common adjustable clock
Pushbuttons	6	Two per axis
10 kΩ resistors	As per schematic	Button pull-ups / input networks
74HC14	1 (shown as U2)	Schmitt-trigger input conditioning
74HC04	Multiple (shown as U1_NOT–U5_NOT)	NOT gates and output inversion
74HC08	3 (shown on schematic)	AND direction-enable logic
74HC193	3	One counter per axis
74HC139	2	Four decoder sections available; three used
ULN2003	3	One driver board / IC per motor
28BYJ-48, 5 V	3	One motor per axis
100 nF ceramic capacitors	One per IC	Local decoupling
Timing resistor, potentiometer and capacitor	As per final 555 design	Sets clock range
Regulated 5 V supply	1 suitable supply	Current capacity must cover all motors and logic
